

Source-Resilient Joint Learning Framework for Preserving Stable Generalization on Diverse Ultrasonic Source Scenarios - Supplementary Materials

I. METHODOLOGY

A. Core Notation Summary

We list the main symbols and variables used throughout the paper:

- x_{tr}, x_{tg} : Ultrasound images from the training source and target source, respectively.
- y^{seg}, y^{cls} : Ground-truth labels for segmentation and classification tasks.
- $\tilde{y}^{seg}, \tilde{y}^{cls}$: Predicted outputs for segmentation and classification.
- $\mathcal{S}_{tr}, \mathcal{S}_{tg}$: The sets of training and pseudo-target samples.
- $f_{\theta}(x)$: Output of the joint model with parameters θ .
- f, f^{aug} : Feature maps before and after augmentation by AFSM.
- $\bar{f}$: Multi-scale fused feature representation generated by TCAM.
- f' : Normalized feature after MCNM projection.
- $\tilde{x}$: Stylized or transformed image via USmapping.
- $T_{src}(\cdot)$: Source transforming operator that simulates the target style.
- $T_{us}(\cdot)$: Ultrasound style randomization function.
- $T_{sr}(\cdot)$: Speckle noise randomization transformation.
- $T_m(\cdot)$: Mask-guided style transfer for anatomical preservation.
- η : Random speckle noise sampled from $\mathcal{U}(0.7, 1.3)$.
- $A(\cdot)$: The AFSM module for stochastic feature augmentation.
- $M(\cdot)$: TCAM module for multi-task consistency.
- $Norm(\cdot; W_n)$: MCNM module that normalizes features using projection matrix W_n .

B. Mathematical Overview of Our Framework

To provide a theoretical foundation for generalization under source heterogeneity in ultrasound, we begin by formulating the learning objective. The main symbols and variables used throughout this paper are presented in *Supplementary Materials*. Conventional generalization theory, such as empirical risk minimization and domain adaptation bounds, suggests that minimizing source loss alone is insufficient to ensure good performance on unseen sources [1]. This motivates the design of our modules. Here, we focus on presenting the high-level theoretical motivations and problem-solution formulations,

while the detailed algorithmic implementations are provided in the subsequent sections.

1) Unified Joint Learning Objective and Generalization Gap:

Let the training data from a single ultrasound source and the test set from a heterogeneous target source be denoted as:

$$\mathcal{S}_{tr} = \{(x_{i,tr}, y_i^{seg}, y_i^{cls})\}_{i=1}^n, \quad \mathcal{S}_{tg} = \{(x_{j,tg}, y_j^{seg}, y_j^{cls})\}_{j=1}^m.$$

We jointly optimize segmentation and classification using:

$$\min_{\theta} \mathbb{E}_{(x, y^{seg}, y^{cls}) \sim \mathcal{S}_{tr}} [\mathcal{L}_{seg}(f_{\theta}(x), y^{seg}) + \mathcal{L}_{cls}(f_{\theta}(x), y^{cls})] \quad (1)$$

The challenge lies in minimizing the generalization gap:

$$\Delta_{gen} = |\mathbb{E}_{x \sim \mathcal{S}_{tr}} \mathcal{L}_{task}(x) - \mathbb{E}_{x \sim \mathcal{S}_{tg}} \mathcal{L}_{task}(x)| \quad (2)$$

The total loss is:

$$\mathcal{L}_{task}(x) = \mathcal{L}_{seg}(f_{\theta}(x), y^{seg}) + \mathcal{L}_{cls}(f_{\theta}(x), y^{cls}) \quad (3)$$

2) Challenge 1: Limited Generalization in Single-Source Training: Problem:

In a single-source setting, the model is trained solely on $\mathcal{S}_{tr}$ and lacks guidance from other sources. As a result, it overfits the source-specific statistics and fails to minimize the generalization gap [2] defined in Eq. (2). Since no distributional shift is observed during training, there is no incentive to learn transferable features, leading to:

$$\mathbb{E}_{x \sim \mathcal{S}_{tg}} \mathcal{L}_{task}(x) > \mathbb{E}_{x \sim \mathcal{S}_{tr}} \mathcal{L}_{task}(x) \quad (4)$$

Our Solution: Source Transforming $T_{src}(\cdot)$. To introduce generalization pressure, we simulate diverse source shifts by transforming each $x_{tr} \in \mathcal{S}_{tr}$ using multiple intensity-based transformations $\mathcal{T}_{intensity}$ [3]. We denote the pseudo-target source generated by T_{src} as $\mathcal{S}_{tg}^{(p)}$. This encourages:

$$\mathbb{E}_{x \sim \mathcal{S}_{tg}^{(p)}} \mathcal{L}_{task}(x) \approx \mathbb{E}_{x \sim \mathcal{S}_{tg}} \mathcal{L}_{task}(x) \quad (5)$$

The details of $T_{src}(\cdot)$ are presented in Eq. (1) of the main paper.

3) Challenge 2: Ultrasound-Specific Style and Noise Biases:

Problem: Ultrasound imaging introduces unique challenges. These include speckle noise, probe-dependent shading, and operator-specific variations, which are hard to eliminate and cause large inter-source shifts even when anatomy is consistent [4]. These corruptions lead to shifts in the feature distribution:

$$\mathbb{E}_{x \sim \mathcal{S}_{tr}}[f(x)] \neq \mathbb{E}_{x \sim \mathcal{S}_{tg}}[f(x)], \quad (6)$$

which violates the assumption of feature-level alignment across sources [5].

Our Solution: USmapping $T_{us}(\cdot)$. To reduce feature distribution gaps while preserving structure, we propose USmapping, which synthesizes stylized samples combined with anatomy-preserving masks. Ideally, USmapping encourages the feature expectations of transformed training samples to approximate those of target samples:

$$\mathbb{E}_{x \sim \mathcal{S}_{tr}}[f(T_{us}(x))] \approx \mathbb{E}_{x \sim \mathcal{S}_{tg}}[f(x)]. \quad (7)$$

The detailed formulation of $T_{us}(\cdot)$ is provided later in Eq. (2) of the main paper.

4) Challenge 3: Inconsistent Feature in Joint Learning:

Problem: In multi-task settings, segmentation and classification tasks often rely on distinct, even conflicting features [6], especially under heterogeneous ultrasound sources. This inconsistency can be formalized as:

$$\mathcal{L}_{joint} = \mathbb{E}_{x \sim \mathcal{S}_{tr}} [\mathcal{L}_{seg}(f^{seg}, y^{seg}) + \mathcal{L}_{cls}(f^{cls}, y^{cls})], \quad (8)$$

where $f^{seg} \neq f^{cls}$, $f^{seg/cls}$ denotes the task-specific features extracted for segmentation/classification. This inconsistency reduces the effectiveness of shared representations and compromises overall generalization.

Our Solution: TCAM $M(\cdot)$. To mitigate this issue, we propose TCAM, a shared attention-based fusion mechanism that integrates multi-scale features and preserves task-level consistency. TCAM aggregates hierarchical features via self-attention to generate a unified representation:

$$\begin{aligned} \mathcal{L}_{joint}^{TCAM} = \mathbb{E}_{x \sim \mathcal{S}_{tr}} [\mathcal{L}_{seg}(M(f), y^{seg}) \\ + \lambda \cdot \mathcal{L}_{cls}(M(f), y^{cls})], \end{aligned} \quad (9)$$

where the detailed formulation of $M(\cdot)$ is presented in Eq. (10) of the main paper.

5) Challenge 4: Limited Feature Diversity in Single-Source Training:

Problem: Training data from a single ultrasound source typically provides limited feature diversity due to fixed acquisition parameters and anatomical similarities. This limitation hinders generalization to unseen sources that may exhibit different imaging settings. Image-level augmentation alone is often insufficient to capture these variations [7]. Formally:

$$\text{Var}(x \sim \mathcal{S}_{tr}) < \text{Var}(x \sim \mathcal{S}_{tg}) \quad (10)$$

Our Solution: AFSM $A(\cdot)$. To address this gap, we introduce a learnable feature-level augmentation that adaptively perturbs training samples. By applying AFSM $A(\cdot)$, the variance of the transformed training distribution approximates that of the target domain:

$$\text{Var}(x \sim A(\mathcal{S}_{tr})) \approx \text{Var}(x \sim \mathcal{S}_{tg}), \quad (11)$$

where the detailed formulation of $A(\cdot)$ is presented in Eqs. (13)–(15) of the main paper.

This feature-level augmentation expands the representation capacity of the model without relying on multiple sources, drawing on principles from distributional robustness [8].

6) Challenge 5: Stabilizing Learning with Manifold Normalization:

Problem: Although AFSM introduces beneficial feature diversity, such stochastic augmentations may cause instability during training if left unconstrained [9]. In particular, augmented features may deviate from the task-relevant geometry:

$$\mathbb{E}_{x \sim \mathcal{S}_{tr}} \mathcal{L}_{task}(A(f)) > \mathbb{E}_{x \sim \mathcal{S}_{tr}} \mathcal{L}_{task}(f), \quad (12)$$

Our Solution: MCNM $Norm(\cdot)$. To mitigate instability, we regularize feature embeddings by minimizing the representation subspace distance (RSD) loss $\mathcal{L}_{RSD}$, a geometry-aware constraint, with its detailed formulation provided in Eqs. (17)–(18) of the main paper. This guides the augmented features back to the task-relevant geometry, and Eq. (12) becomes approximately balanced:

$$\begin{aligned} \mathbb{E}_{x \sim \mathcal{S}_{tr}} \mathcal{L}_{task}(Norm(A(f))) \approx \\ \mathbb{E}_{x \sim \mathcal{S}_{tr}} \mathcal{L}_{task}(Norm(f)). \end{aligned} \quad (13)$$

C. Algorithmic Implementation

The algorithmic implementations of AFSM, TCAM, and MCNM are detailed in Algorithm 1, 2, and 3, respectively.

Algorithm 1 Adaptive Feature-Shifting Module (AFSM)

Input: Feature map $f \in \mathbb{R}^{B \times C \times H \times W}$, learnable parameters

$\mu_{l1}, \sigma_{l1}, \mu_{l2}, \sigma_{l2}$

Output: Augmented feature $f^{aug} \in \mathbb{R}^{B \times C \times H \times W}$

- 1: Compute $\mu_f = \text{Mean}(f)$
 - 2: Compute $\sigma_f = \text{Std}(f)$
 - 3: Sample $\epsilon_1 \sim \mathcal{N}(\mu_{l1}, \sigma_{l1})$, $\epsilon_2 \sim \mathcal{N}(\mu_{l2}, \sigma_{l2})$
 - 4: Sample $p_1, p_2 \sim \text{Bernoulli}(0.5)$
 - 5: Compute $\gamma = \sigma_f + p_1 \cdot \epsilon_1$
 - 6: Compute $\beta = \mu_f + p_2 \cdot \epsilon_2$
 - 7: Compute $f^{aug} = \gamma \cdot \frac{f - \mu_f}{\sigma_f + \epsilon} + \beta$
 - 8: **return** f^{aug}
-

Algorithm 2 Task-consistent attention module (TCAM)

Input: Multi-scale features $\{f_n^{aug}\}_{n=1}^N \in \mathbb{R}^{B \times C \times H_n \times W_n}$

Output: Fused feature $\bar{f} \in \mathbb{R}^{B \times C \times H \times W}$

- 1: **for** $n \leftarrow 1$ **to** N **do**
 - 2: Resize f_n^{aug} to target size (H, W) as f_n^{res}
 - 3: Compute: $Q_n = W_q f_n^{res}$, $K_n = W_k f_n^{res}$, $V_n = W_v f_n^{res}$
 - 4: Compute attention: $\text{Attn}_n = \text{Softmax}\left(\frac{Q_n K_n^T}{\sqrt{C}}\right) V_n$
 - 5: **end for**
 - 6: Concatenate $\{\text{Attn}_n\}_{n=1}^N$ along channel dimension
 - 7: Fuse features: $\bar{f} = \text{Conv}_{1 \times 1}(\text{Concat}(\text{Attn}_1, \dots, \text{Attn}_N))$
 - 8: **return** $\bar{f}$
-

Algorithm 3 Manifold-constraint normalization module (MCNM)

Input: Fused feature $\bar{f} \in \mathbb{R}^{B \times C \times H \times W}$, Learnable projection matrix $W_n \in \mathbb{R}^{(C \cdot H \cdot W) \times d}$

Output: Normalized feature $f' \in \mathbb{R}^{B \times C \times H \times W}$

- 1: Flatten: $\bar{F} = \text{Flatten}(\bar{f})$
- 2: Project: $F_{\text{proj}} = \bar{F}W_n$
- 3: Compute SVD: $U_{tr}, \Sigma_{tr}, V_{tr}^\top = \text{SVD}(F_{\text{proj}}^{tr})$
- 4: Compute SVD: $U_{tg}, \Sigma_{tg}, V_{tg}^\top = \text{SVD}(F_{\text{proj}}^{tg})$
- 5: Compute principal angles: $\theta_i = \arccos(\sigma_i(U_{tr}^\top U_{tg}))$
- 6: Compute RSD loss: $\mathcal{L}_{RSD} = \sum_{i=1}^p \sin(\theta_i)$
- 7: Update W_n : $W_n \leftarrow W_n - \eta \nabla_{W_n} \mathcal{L}_{RSD}$
- 8: Reshape F_{proj} to f'
- 9: **return** f'

D. Pipeline of USmapping

Fig. S1 presents the pipeline of our proposed USmapping module. The USmapping consists of two stages: speckle randomization (SpeckleRand) and mask-guiding Monge-Kantorovitch linear (MaskMKL). SpeckleRand applies speckle reduction and adds speckle noise randomly to incorporate the network with knowledge of speckle noise. Then, MaskMKL is utilized to embed the multi-source key information into the network, leveraging the strengths of conventional style transfer, mask-guiding style transfer, and Gaussian image fusion. The key region for liver-related tasks is the liver parenchyma, while for nodule-related tasks, it is the nodule.

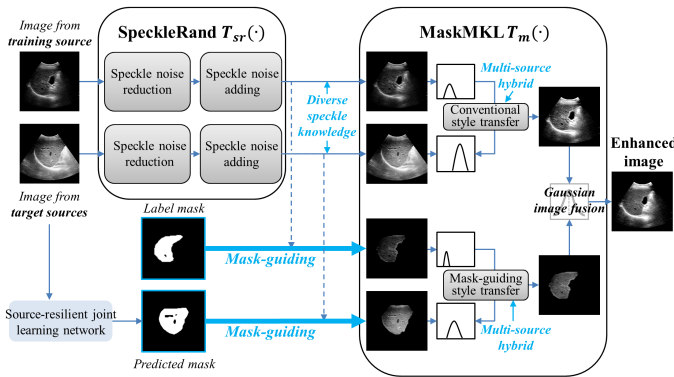


Fig. S1. Pipeline of our proposed US-hybrid linear mapping (USmapping).

E. Stably Training Strategy of MCNM

Fig. S2 shows the training process of our proposed MCNM. Firstly, the images of training source x_{tr} and target sources x_{tg} are fed into the source-resilient joint learning network to obtain reference features f'_{tr} and normalized features f'_{tg} , respectively. Then, $\mathcal{L}_{RSD}$ is computed between f'_{tr} and f'_{tg} . The gradient for $\mathcal{L}_{RSD}$ only updates the parameters W_n of the MCNM. By using the MCNM, the features from different sources are normalized to a shared feature space, improving the model performance on different sources.

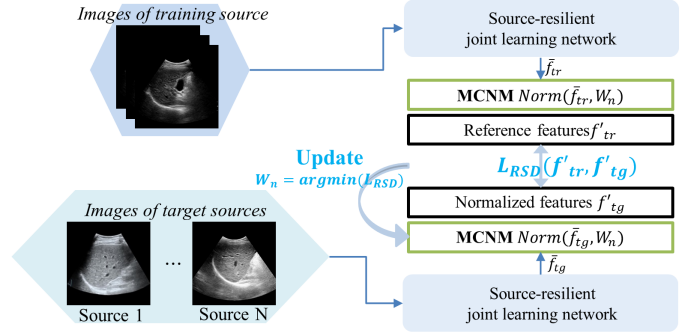


Fig. S2. Stably training strategy of manifold-constraint normalization module (MCNM).

F. Implementation details

We adopted Pytorch to implement our framework, which was trained on Nvidia RTX 3070 with 8 GB memory. EfficientNet-B1 was used as backbone due to its fewer parameters and good performance in most tasks. A network with fewer parameters is benefit to prevent over-fitting.

In the training process, the training batch size was set to 4, the maximum epoch number was 150, and the initial learning rate was set to 0.0005. The learning rate was decayed with a cosine annealing for each epoch, and the minimum learning rate was set to 0.000001. The restart epoch of cosine annealing was set to 150. It indicated that the learning rate monotonically decreased in the training process. In our experiments, the optimizer was AdamW [10]. The weight decay, β_1 , and β_2 were set to 0.01, 0.9, and 0.999, respectively.

Several traditional transformations are employed in our framework for traditional augmentation to prevent over-fitting, including gamma correction ranging from 0.5 to 1.5, brightness adjustment ranging from -0.2 to 0.2, image flipping, image rotation ranging from -45° to 45° , image scaling ranging from 0.6 to 1.4, and deformable transformation. In the comparison experiments and ablation studies, all models were trained by using this traditional augmentation method.

II. EXPERIMENTAL RESULTS

A. Datasets

We did not use the data categorized as normal in the BUSI dataset because all data in other datasets at least contain a lesion. In the OASBUD dataset, only radio frequency signals are provided, and thus we reconstructed the ultrasound image using the method in [11]. For a fair comparison, all experiments were performed based on the same splitting. The results of all experiments are computed on the testing set of each source. In the data preprocessing, the input images were resized to 224×224 and were divided by 255 for normalization.

Private dataset labels were reliable. For segmentation tasks, liver and thyroid nodule regions were manually delineated by an expert and verified by a sonographer. For classification tasks, liver fibrosis was staged on a five-level scale based on histopathology reports. Stages F0 and F1 were treated as negative, and F2 to F4 as positive. Thyroid nodules were

diagnosed as benign or malignant, with benign as negative and malignant as positive samples.

B. Evaluation metrics

We weighted average the evaluation metrics across the datasets for more reasonable comparisons. Since the datasets used in our experiments vary significantly in sample size, it would be unfair to directly average these metrics for comparison. Thus, in addition to direct averaging, we weighted average the evaluation metrics according to the sample size of each dataset, and the formulas are as follows:

$$DSC_{W Avg} = \frac{\sum_{i=1}^N DSC_i \cdot w_i}{\sum_{i=1}^N w_i} \times 100\%, \quad (14)$$

$$AUC_{W Avg} = \frac{\sum_{i=1}^N AUC_i \cdot w_i}{\sum_{i=1}^N w_i} \times 100\%, \quad (15)$$

where $DSC/AUC_{W Avg}$ denotes the weighted average of DSC/AUC across datasets, DSC/AUC_i denotes the DSC/AUC of dataset i , N denotes the number of the dataset, and w_i denotes the weight of dataset i , which is the sample number of the corresponding test set.

C. Results of Computational Complexity Analysis

Table SI presents the inference and training times, as well as the model parameters, for different module configurations.

TABLE SI
COMPUTATIONAL COMPLEXITY ANALYSIS OF TCAM, AFSM, MCNM, AND USMAPPING IN TERMS OF INFERENCE TIME, TRAINING TIME AND MODE PARAMETERS.

TCAM	AFSM	MCNM	USmapping	Inference (s)	Training (s)	Parameters
-	-	-	-	0.016	0.017	8.8M
✓	-	-	-	0.030	0.035	24.8M
✓	✓	-	-	0.031	0.036	24.8M
✓	✓	✓	-	0.032	0.037	25.3M
✓	✓	✓	✓	0.032	0.302	25.3M

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